

Joint NTN Slicing and Admission Control for Infrastructure-as-a-Service: A Deep Learning Aided Multi-Objective Optimization

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Abstract—Recently, there has been a surge in the adoption of multi-orbital satellite networks for integrated service delivery. Operators are increasingly collaborating and constructing multi-layer network infrastructures to meet growing traffic demands. This paper introduces a novel service delivery model where infrastructure providers (InPs) lease out resources from non-terrestrial networks (NTNs) as slices to mobile virtual service operators (MVSOs). These MVSOs then offer the leased resources to subscribers, facilitating efficient utilization of NTN resources in the telecommunications ecosystem. The model utilizes an innovative NTN slicing architecture that incorporates multi-layer satellites, including low Earth orbit (LEO), medium Earth orbit (MEO), and geostationary orbit (GEO) constellations. It features a hybrid gateway station (HGS) tailored to the virtualization architecture specified by the 3rd Generation Partnership Project (3GPP). In this setting, we formulate a multi-objective optimization problem (MOOP) comprising two combinatorial objective functions for InPs and MVSOs, aiming to maximize revenue. The proposed algorithm addresses the joint network slicing and admission control (AC) requirements by employing techniques such as non-dominated sorting genetic algorithm II (NSGA-II), multi-objective reinforcement learning (MORL), and a heuristic approach. Our algorithm outperforms the Round Robin (RR) and Max-Min fairness approaches, achieving increases in peak revenue of 3.91% and 18.73%, respectively.

Index Terms—Network slicing, deep learning, admission control, multi-connectivity, non-terrestrial network (NTN), resource allocation, satellite communications, MORL, LSTM.

I. INTRODUCTION

THE EMERGENCE of digital innovations and new technologies, such as the Internet of Things (IoT), machine-type communications, and Industry 4.0 solutions,

has significantly transformed the demands for connectivity in today's world [1]. As a result, there is a growing need for reliable and widespread global connectivity. Satellite communication systems offer a promising solution to address these requirements, as they can provide coverage in remote areas, enable seamless connectivity across vast distances, and support a wide range of applications [2]. With their ability to overcome geographical limitations and reach under-served regions, satellite communication systems have the potential to meet the evolving connectivity needs of various industries and sectors. In this context, the specifications outlined by the 3rd Generation Partnership Project (3GPP) regarding satellite connectivity have created an intriguing prospect for terrestrial communication service providers and satellite operators to collaborate symbiotically and establish a global non-terrestrial network (NTN) ecosystem [3]. Specifically, 3GPP has made efforts to adapt the fifth-generation (5G) New Radio (NR) as well as narrow-band IoT (NB-IoT) and long-term evolution (LTE) for machine-type communications (LTE-M) to provide satellite-based connectivity [4]. This NTN ecosystem is expected to bring significant benefits to individuals and organizations by providing global connectivity opportunities [5].

The emerging use cases in wireless connectivity necessitate a departure from the conventional one-size-fits-all approach, as the diversity of service characteristics among these new use cases continues to expand. For instance, Industry 4.0 seeks flexible and highly customizable solutions, creating a demand for rapid deployment, configuration, modification, and cost-effective wide-area coverage. In the past, private networks were employed to address specific communication requirements of user groups, but these networks often had limited options for connection quality, functionality, and customization. To overcome these issues, 5G technology introduces network slicing as an effective solution. Network slicing enables the partitioning of wireless networks into virtual network elements or *slices* that can be configured to suit the unique needs of applications, services, devices, customers, or operators. Each network slice operates as a form of private network, running on a shared physical infrastructure while maintaining specified quality of service (QoS) requirements. This approach combines the benefits of private networks with the efficiency and scalability of a public network.

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Additionally, data traffic in non-terrestrial networks (NTNs) is diverse and randomly distributed in the serving areas, and coming from various users with different QoS requirements [6]. Thus, employing network slicing in NTNs would also help satisfying the asymmetry and heterogeneity of the traffic demands. Specifically, NTN slicing allows for the efficient allocation of resources and tailored services to different user segments. For instance, it enables the separation of traffic between critical applications like emergency services and the less time-sensitive data transfers, optimizing system capacity utilization and ensuring uninterrupted connectivity in critical situations. NTN slicing also enhances security by isolating different user groups, minimizing the risk of data breaches or interference. Further, NTN slicing can foster innovation by enabling experimentation with new services and applications within isolated slices, without affecting the entire network.

A. Related Works

In the literature, several contributions have studied network slicing for terrestrial communications, including 5G, and some have integrated machine learning techniques to augment and streamline resource management processes. In [7], terrestrial 5G slicing is presented with a forecasting scheme per slice, employing an admission control decision process that considers traffic and spatial information to meet the required service level agreements (SLAs). Additionally, reinforcement learning (RL) has been integrated into the system to enhance overall performance. Q-learning, a form of RL, is applied in the 5G network slicing framework in [8], which was analyzed with admission and congestion control mechanisms.

Similarly, a Q-learning aided algorithm has been developed in [9] to maximize revenue for the infrastructure providers (InPs) while ensuring QoS as required by the mobile virtual service operators (MVSOs), who lease network capacity in the form of slices from the InPs. This model is known as Infrastructure-as-a-Service (IaaS). The authors in [10] explored slice overbooking to maximize the terrestrial operator's revenue in an end-to-end orchestrated slicing framework. A radio access network (RAN) domain slicing for 5G is studied in [11], where a slice admission strategy based on RL is presented. The service demands have different priorities, upon which the InPs make decisions on slice requests to accept or reject in order to maximize revenue. Similarly, authors in [12] and [13] present blockchain-based resource trading between InPs and mobile virtual network operators, aided with RL for 5G RAN slice resource allocation.

In addition, a RL algorithm is presented in [14], which defines a policy that jointly optimizes admission and congestion control to maximize system reward while handling prioritized slices. A type of deep learning (DL) prediction algorithm, known as long short-term memory (LSTM), is studied in [15] and [16] to predict traffic demand requirements for efficient resource allocation in 5G network slicing, in order to avoid revenue loss due to forecasting errors. A joint network slicing and admission control (AC) problem has been formulated and solved using multi-agent deep RL in [17] for revenue maximization, considering three classes of

slice requests: enhanced Mobile Broadband (eMBB), massive Machine-Type Communication (mMTC) and Ultra-Reliable Low-Latency Communications (URLLC) with real dataset.

Further, in satellite communications, research on network slicing has mainly focused on single-layer low Earth orbit (LEO) constellations [18], [19], overlooking the significant potential of incorporating multi-layered NTN architectures, which could harness the added resilience and reliability inherent in multi-orbit satellite systems [20]. This slicing paradigm has the potential for dynamic QoS offerings and to satisfy diverse capacity demands. In addition, integrating multi connectivity (MC) into the network slicing framework offers several advantages for the multi-layer NTNs. Specifically, it enables seamless load balancing among satellite layers, optimizing resource use and ensuring consistent service quality across orbits. MC also enhances network redundancy, reducing the impact of disruptions and improving reliability [21].

These observations have motivated this work to combine multi-layer NTN slicing with MC to further enhance the system capacity and communications reliability. Interestingly, multi-layer NTN has attracted industry interest as many operators are investing in expanding their network portfolios with additional orbital constellations or merging with other operators that offer different orbital satellite capacities, for instance SES operates medium Earth orbit/geostationary orbit (MEO/GEO) constellations [22], and the merger of Eutelsat with Oneweb occurred for LEO/GEO service offerings [23]. However, the amalgamation of NTN slicing and MC brings about multiple challenges due to the inherent complexity and the dynamic nature of the NTN entities in order to realize the benefits fully. This intricate scenario underscores the significance of incorporating machine learning as a valuable tool within this framework, given its capacity to handle complex and non-linear time-series features within the data, especially for prediction.

In this complex setup, multi-objective optimization is essential to consider multiple, often conflicting, objectives simultaneously when making decisions. In our scenario, it is insufficient to optimize for just one objective because there are multiple objectives that need to be balanced or traded off against each other. In this context, multi-objective optimization problems (MOOPs) provide a framework for systematically exploring the trade-offs between these objectives and finding optimal solutions that balance them effectively. For example, the authors in [24] explored a form of genetic algorithms (GA) known as non-dominated sorting genetic algorithm II (NSGA-II) to solve a two-objective multi-objective optimization problem (MOOP) involving the maximization of profit and the minimization of satellite maneuver angle in GEO satellites. Further, RL-aided approach was considered in [25] to solve MOOP involving three objectives for dynamic beam hopping in satellites, using an algorithm labeled as multi-objective reinforcement learning (MORL).

B. Contributions

To this end, we formulate a MOOP and propose a novel algorithm that utilizes GA and MORL, along with deep

TABLE I
COMPARISON OF RELATED LITERATURE ON NETWORK SLICING AND ADMISSION CONTROL

Main Topics	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]	[19]	This work
Terrestrial Network slicing	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
NTN Network slicing												✓	✓	✓
Multi-layer NTN Network slicing														✓
Block-chain based secure slicing						✓	✓							
Admission control with priority		✓			✓			✓	✓	✓	✓			✓
Congestion control		✓						✓						
Problem formed as MOOP						✓	✓							✓
Problem formed as non-MOOP	✓		✓	✓	✓			✓	✓	✓	✓	✓	✓	
Optimization with genetic algorithm		✓		✓									✓	✓
Optimization with machine learning	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓		✓
Synthetic dataset									✓					
Real dataset										✓	✓			✓
Traffic prediction/forecasting	✓			✓					✓	✓				✓
Optimization with Heuristic	✓			✓										✓
Revenue maximization objective	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓
QoS satisfaction objective		✓	✓			✓	✓					✓	✓	✓
Efficient resource utilization	✓	✓				✓		✓		✓				✓
SLA violation minimization	✓	✓							✓					✓

learning predictions implemented with realistic dataset inputs, to enhance AC decisions while adapting to traffic fluctuations, to minimize service level agreement (SLA) violations and maximize revenue within the network.

Contributions: Our key technical contributions can be explicitly summarized as follows:

- 1) Design a novel network slicing architecture for a multi-layer NTN topology comprising of GEO/MEO/LEO satellites and hybrid gateway station (HGS) [26]. This architecture is adapted to the 3GPP specification with network orchestration management flow, incorporating machine learning applications and resource scheduling integration.
- 2) Provide a practical business model framework for functional service operations, which entails mobile virtual service operator (MVSO) leasing satellite infrastructural network resources involving a multi-layer NTN. This model aims to offer services to end-users based on agreed SLAs.
- 3) Develop a joint network slicing and AC algorithm that considers dynamic traffic demand involving eMBB and URLLC, along with the available network resources with QoS and SLA requirements. The algorithm is designed from a MOOP to maximize the revenue of the infrastructure provider (InP) and the MVSO over some constraints. This problem is solved using three approaches: the first involves optimal vectors obtained from a Pareto curve using NSGA-II, the second utilizes MORL, and the last employs a Heuristic approach.
- 4) Introduce a novel AC mechanism known as revenue maximization (Rev-Max) algorithm, which is aided by recurrent neural networks (RNNs) using LSTM algorithm, for predicting the dynamic traffic demands of the two classes, eMBB and URLLC, which have diverse radio frequency features. The AC mechanism considers the available capacities of divergent LEO, MEO and GEO satellites, including the different slice weight of the demanded traffic classes and the priority configured

for each class. It then admits the slices cognitively while ensuring that QoS requirements are satisfied.

The rest of the paper is structured as follows. In Section II, the network model and architecture are discussed in detail. Section III focuses on the multi-objective optimization problem, while the optimization solution and the proposed algorithms are outlined in Section IV. In Section V, the simulation setup and performance evaluation are discussed. Finally, the conclusion is summarised in Section VI.

II. NETWORK MODEL AND ARCHITECTURE

A. Service Flow Model

The service and business flow model is depicted in Fig. 1, which involves three actors (i) the end user (subscribers), (ii) MVSOs, and (iii) InPs. The InP owns the network infrastructure over which services are offered by the MVSOs. This infrastructure involves satellite RAN connected to the core network (CN). Both hardware and virtualized/softwarized components of the network are leased as network resource offerings to multiple MVSOs for a fee. Details are discussed in Section II-B. The MVSOs offer services such as broadband connectivity to cruise ships, trains, and airplanes, as well as multicast/broadcast content to homes. Additionally, some telecom operators requiring satellite backhauling services can also be categorized as MVSOs. The MVSOs provide subscribers with different QoS options based on their needs and budget, enabling access to a variety of broadband services accompanied by SLAs.

B. Network Topology and Architectural Framework

1) *Multi-Domain Network:* The network topology consists of RAN connected to a CN domain through the HGS. The RAN domain comprises the three orbital layers: LEO, MEO and GEO satellite constellations. These constellations are linked to a pool of interconnected hybrid gateway stations, and a common network operation center (NOC) for management and control, resulting in a multi-layer NTN RAN.

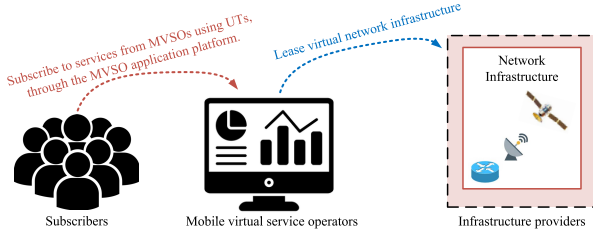


Fig. 1. Service and business flow model involving subscribers, virtual operators and infrastructure providers.

2) *Network Slicing*: Parts of the multi-layer NTN RAN have been adapted into the 3GPP network function virtualization (NFV) reference architecture [27], where network virtualization and softwarization are implemented for effective resource slicing and orchestration, as shown in Fig 2. Network slicing enables the dimensioning of network resources to cater to different individuals, organisations or third-party service suppliers/operators simultaneously. This technology has become an essential part of 5G standardization, presenting a business case where InPs can maximally utilize their resources for optimum revenue generation. The network topology shows transmission sessions for multiple users with different traffic classes of URLLC and eMBB, denoted as ω_1 and ω_2 , respectively. These users subscribe to the various services offered by MVSOs. To meet the dynamic lease demands of different MVSOs, the InP must dimension the virtual components of the physical network into portions known as network slices. These lease demands arise as slice requests (SRs) from different MVSOs. Each SR is accompanied by the desired QoS specifications and RAN/CN capacity requirements. Further, it is important to note that ensuring reliability and real-time performance in network slicing involves isolating slices to prevent interference, dynamically allocating resources, and managing slice life-cycles through orchestration. Implementing QoS policies and NFV enhances performance and scalability. The enforcement of SLAs, and incorporating redundancy also ensures reliability, while automation reduces errors and improves response times. The solution proposed in this paper is designed to ensure reliability and real-time performance.

3) *Admission Control*: The AC for the SR is executed in the HGS. The indicator of the accepted SR, α_k , is sent to the HGS controller so that SR k can be admitted into the RAN. This mechanism evaluates the total lease capacity from the InP, to ensure adequate accommodation of MVSO slice demand based on the slice request (S_w)¹, and the corresponding constraints on the MVSO SLAs (additional details in Section III). This is performed to establish an acceptance policy for all SR, based on the defined objective of the problem. In the current setup, revenue maximization is considered as an objective, with details provided in Sections III and IV. Subsequently, appropriate RAN resources are allocated to ensure that the SLA is not violated, based on the dynamic traffic insights

obtained using a deep learning model of RNN, details of which are deferred to Section IV-C.

C. Downlink Transmission Model

The transmission involves datacenter/CN to end users through the RAN domain. This paper focuses on transmission in the RAN domain from the HGS over the multi-layer NTN and satellite channel, to the end users.

1) *Link Budget Model*: The transmission link is established based on link budget considerations to achieve reliable transmission over the channel. For a satellite link, the overall propagation loss (L_{Total}) in dB between the satellite and the user terminal (UT) is:

$$L_{Total} = L_{FSL} + L_{atm} + L_{\zeta} + L_{sc} \quad (1)$$

where L_{atm} is the atmospheric loss as provided in ITU-R P.676 [28], and L_{sc} is the loss from tropospheric scintillation, which affects Ka band signals, with values obtained as described in ITU-R P.618 [29]. L_{ζ} stands for the random lognormal variable with a variance based on the shadowing environment and zero mean, i.e., $L_{\zeta} \sim (0, \zeta^2)$, with 3GPP providing the values of ζ^2 ; this term is influenced by the elevation angle for different cases of rural, dense urban, and urban environments [30]. The free space loss is represented as L_{FSL} in (2), which is a function of the propagation distance between the satellite and the UT.

$$L_{FSL} = 32.45 + 20 \log_{10} F_r + 20 \log_{10} d_p \quad (2)$$

where d_p (in meters) is the propagation distance, and F_r (in GHz) is the carrier frequency. Assuming a downlink satellite communications network where UTs are scattered over different satellite beams, there exist intended user UT_n randomly positioned within the coverage area of the intended satellite S_s , while satellite S_a transmitting to user UT_a interferes with UT_n . The intended receive power by UT_n from satellite S_s is

$$C_{s,n} = EIRP_s + G_n - L_{Total} \quad (3)$$

where G_n is the gain of the receiving UT_n , and $EIRP_s$ is the downlink equivalent isotropically radiated power from the intended satellite S_s towards intended UT_n . The noise power observed at the receiver is computed as:

$$N = N_{fig} + 10 \log_{10} \left(T_0 + (T_{ant} - T_0) 10^{-0.1 N_{fig}} \right) + K_B + 10 \log_{10}(B) \quad (4)$$

where N_{fig} is the noise figure at the receiver, T_0 is the reference noise temperature (290 K), T_{ant} is the temperature of the receiving antenna, K_B is the Boltzmann's constant in dBW/K/Hz, and B is the bandwidth in Hz. Further, the corresponding downlink carrier to noise ratio can be obtained in dB by subtracting (4) from (3), as shown in (5):

$$\left(\frac{C}{N} \right)_{s,n} = EIRP_s + G_n - L_{Total} - N \quad (5)$$

¹ A S_w is the unit of slice request per individual MVSO request, out of the total SR in that traffic category.

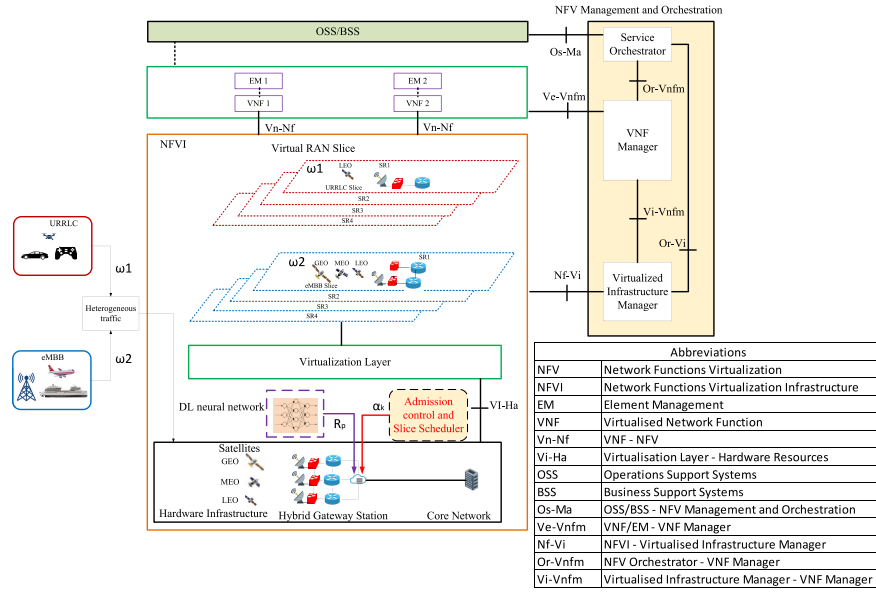


Fig. 2. Schematic diagram of the considered DL-aided network slicing architecture over the topology of the Multi-layer NTN.

The total interference captured at the intended receiving UT_n in the downlink is derived as I_n in dB, as shown below:

$$I_n = 10 \log_{10} \left(\sum_{ch=1}^{N_{ch}} 10^{0.1 I_{a,n}} \right) \quad (6)$$

where $I_{a,n}$ is the interference from the a -th interfering satellite/beam, and $ch = 1, \dots, N_{ch}$, with N_{ch} being the number of co-channel beams.

$$I_{a,n} = EIRP_a + G_n - L_{Total} \quad (7)$$

The carrier to interference ratio in dB is obtain by subtracting (6) from (3)

$$C/I_{s,n} = EIRP_s + G_n - L_{Total} - I_n \quad (8)$$

Hence, the carrier to interference with noise ratio (C/IN) in dB is given below from [30]:

$$\gamma_{s,n} = \frac{C}{IN_{s,n}} = -10 \log_{10} \left(10^{-0.1 \frac{C}{N_{s,n}}} + 10^{-0.1 \frac{C}{I_{s,n}}} \right) \quad (9)$$

2) *Achievable Rate*: The achievable rate on the forward transmission link from satellite S_s to receiver UT_n , utilizing bandwidth B summed over number of intra/inter orbital satellite component carriers C is given as:

$$R_n = \sum_{c=1}^C B \log_2(1 + \gamma_{s,n}) \quad (10)$$

The sum rate for each provision slice k is derived below:

$$R_k = \sum_{n=1}^{N_k^u} R_n \quad (11)$$

where N_k^u is the total number of users in slice k .

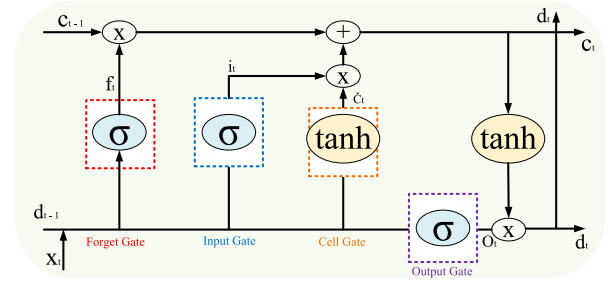


Fig. 3. LSTM cell architecture.

D. Deep Learning Traffic Prediction Model

In this paper, a RNN-based deep learning model is adopted since it can solve time-series learning problems. To tackle the vanishing gradient challenge in RNN, LSTM neural network was proposed in [31] with memory cell, and additional improvement was introduced with the addition of gates in [32]. The LSTM cell architecture is presented in Fig. 3 [33], where the forget gate (f_t), input gate (i_t), cell gate (C_t), and the output gate (O_t) are shown. Through these gate functions, LSTM can capture difficult features in time-series for either short or long terms [34]. To handle temporal dependencies, LSTM defines and maintains the cell state to regulate information flow over time, a vital mechanism of the LSTM structure [35]. To effectively address long-term and short-term dependencies in time-series data from satellite communications applications, the LSTM architecture is a suitable prediction solution [36].

The LSTM architecture performs the following actions discussed further. Firstly, the LSTM categorize the input data as important and redundant, crucial to eliminating redundant information irrelevant for data prediction. The forget gate decides which information should be discarded or retained by the LSTM, and this is shown below from [37].

$$f_t = \sigma(W_{f,t}x_t + W'_{f,t}d_{t-1} + b_{f,t}) \quad (12)$$

where sigmoid function is denoted as σ , and the forget weight in time t are $W_{f,t} \in \mathbb{R}^{Z \times Y_{in}}$ and $W'_{f,t} \in \mathbb{R}^{Z \times Z}$, while the biases at time t are represented as $b_{f,t} \in \mathbb{R}^{Z \times 1}$. Further, $x_t \in \mathbb{R}^{Y_{in} \times 1}$ denotes the input vector of size Y_{in} and d_{t-1} is the previous hidden state of size Z . The second step is for the LSTM cell to apply computations on the information through the input gate to store the relevant information as:

$$\begin{aligned} i_t &= \sigma(W_{i,t}x_t + W'_{i,t}d_{t-1} + b_{i,t}) \\ \bar{c}_t &= \tanh(W_{\bar{c},t}x_t + W'_{\bar{c},t}d_{t-1} + b_{\bar{c},t}) \end{aligned} \quad (13)$$

The third step involves LSTM updating the current cell state c_t with the previously mentioned two steps above.

$$c_t = f_t \odot c_{t-1} + i_t \odot \bar{c}_t \quad (14)$$

where $\odot$ is the Hadamard product.

The fourth and final step is to generate the LSTM cell output. This is achieved by updating the hidden state and generating the output using the output gate, as shown in the computation below:

$$\begin{aligned} o_t &= \sigma(W_{o,t}x_t + W'_{o,t}d_{t-1} + b_{o,t}) \\ d_t &= o_t \odot \tanh c_t \end{aligned} \quad (15)$$

The metric evaluating the effectiveness of the model is the mean absolute error (MAE) [38] given by:

$$MAE = \frac{1}{T} \sum_{t=1}^T |R_a - \widehat{R}_p| \quad (16)$$

where T is the duration of prediction, R_a and $\widehat{R}_p$ are the actual and predicted traffic, respectively.

E. Datasets

The dataset comprises averaged real measured satellite traffic data from the operator SES networks, which has been preprocessed into two traffic classes with different QoS specifications, denoted as ω_1 and ω_2 for URLLC and eMBB, respectively. The data represents hourly traffic demand for a period of 22 days. For URLLC, the service specifications include latency ranging from 10 ms to 50 ms, throughput from 20 Mbps to 100 Mbps, and availability of 99% [39]. The corresponding use-cases for delay-intolerant communications include factory automation, airborne base stations for non-public network (NPN), autonomous driving, autonomous ship, and medical imaging/video traffic [40]. On the other hand, eMBB service specifications require 150 ms to 300 ms latency, 35 Mbps to 1,200 Mbps throughput, and 99% availability [41], [42], [43], catering to use-cases such as 5G backhauling and multimedia content delivery. Hence, the dataset aligns with the outlined specifications, such that ω_1 and ω_2 operate with a single-user maximum guaranteed bit rate (GBR) of 47 Mbps and 115 Mbps, respectively.

III. THE OPTIMIZATION PROBLEM

This problem involves both slicing and AC, aiming to maximize operator revenue while ensuring efficient resource utilization and maintaining QoS requirements. Two functions are considered to create a multi-objective function.

A. Infrastructure Objective Function

The objective of this problem is to maximize revenue for the InP while ensuring that QoS requirements are satisfied and SLAs are not violated for the MVSOs.

$$F_i = \sum_{k=1}^K \sum_{s=1}^S \alpha_k \vartheta_k^s R_k^s \beta_{k,s} - \zeta \vartheta_k^s R_k^s \quad (17)$$

where α_k is the indicator variable for the acceptance of the k -th slice, ϑ_k^s is the offered price by the InP in terms of data rate as \$/Mbps unit, R_k^s is the cumulative data rate of slice k in satellite s in Mbps unit, $\beta_{k,s}$ indicates a binary resource association indicator where k -th slice is allocated to satellite bandwidth resource on satellite s . The overhead cost incurred by the InP for leasing capacity for slice k , associated with satellite s , is represented as $\zeta \vartheta_k^s R_k^s$, where ζ is the InP overhead cost constant that satisfies $0 \leq \zeta \leq 1$.

The InP optimization problem is formulated as follows:

$$\text{maximize}_{\vartheta_k^s, \beta_{k,s}} F_i \quad (18)$$

subject to $C_1 - C_6$

The first constraint C_1 indicates the acceptance and admission of network SR k as α_k . Priority consciousness is configured where each SR is either of higher priority (H-P) or lower priority (L-P). SR priority is proportional to the maximum GBR of the slice, captured in the SR traffic class as ω . This provides identification for the types of resources that are accepted and rejected.

$$C_1 : \alpha_k = \begin{cases} 1, & \text{if slice is accepted;} \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

Constraint C_2 ensures that the required RAN capacity in terms of total bandwidth B_k^s in MHz for all accepted slices α_k , is not more than the available satellite bandwidth $\bar{B}_s$ in MHz of all satellites S . As a result, the bandwidth resources are not over-dimensioned per SR, allowing for the acceptance of additional SR into the network.

$$C_2 : \sum_{k=1}^K \alpha_k B_k^s \leq \sum_{s=1}^S \bar{B}_s \quad (20)$$

Constraint C_3 ensures that the data rate R_k of slice k lies between a maximum (R_ω^{max}) and minimum (R_ω^{min}) limit, based on the traffic class ω of either URLLC or eMBB as agreed upon in the SLA.

$$C_3 : \alpha_k R_\omega^{min} \leq \beta_{k,s} \cdot R_k^s \leq \alpha_k R_\omega^{max} \quad \forall k \in K \quad \forall s \in S \quad (21)$$

The constraint on the end-to-end (E2E) delay limit requirement for slice k on satellite s is given in C_4 . It also ensures that E2E delay D_k in ms, as per SLA, is guaranteed.

$$C_4 : D_k \beta_{k,s} \leq D_k^{max} \quad \forall k \in K \quad \forall s \in S \quad (22)$$

In this problem, the different slices $\forall k \in K$ have different traffic categories with varying QoS requirements, involving throughput and delay.

$$C_5 : \sum_{s=1}^S \beta_{k,s} \leq \Omega \quad \forall k \in K \quad (23)$$

Constraints C_5 restricts the accepted slice α_k to utilize bandwidth resources from a maximum of Ω number of inter/intra orbital satellites, facilitating MC for user terminals. This allows the slice k to associate with only a limited number of satellites in RAN, preventing congestion or association rejection for other slice request.

$$C_6 : D_k^{max} = \begin{cases} \max\{D_k^{LEO}\} + T_{sig} & \text{if } \omega = URLLC \\ \max\{D_k^{GEO}, D_k^{MEO}, D_k^{LEO}\} + T_{sig} & \text{if } \omega = eMBB \\ 0 & \text{otherwise} \end{cases} \quad (24)$$

Constraint C_6 further defines the actual maximum delay mapping for the traffic classes URLLC or eMBB, with respect to orbital satellite required to serve the traffic class in question. Here, $D_k^s(ms) = \lceil \frac{d_k^s}{c} \rceil$ [44], where d_k^s (in m) is the propagation distance through satellite s (LEO, MEO or GEO), and c is the speed of light in m/ms . Additionally, T_{sig} is the signal processing time in ms.

This is a non-convex combinatorial optimization problem with inequality constraints, which maximizes the variables ϑ_k^s and $\beta_{k,s}$.

B. MVSO Objective Function

The objective of this problem is to maximize MVSO revenue by enhancing the subscriber satisfaction index (ρ_n) for all users N , which will motivate enhanced service patronage. The subscriber satisfaction index is a performance indicator determined by the user data rate, delay, number of users and unit price; it increases when rate increases and the delay reduces, among other factors. The function in (25) captures the net revenue of the MVSO, where the cost of leasing the infrastructure is deducted as $\vartheta_k^s R_{n,k}^u$, and gain from satisfaction index is presented as ρ_n^2 .

$$\bar{F}_m = \sum_{n=1}^N \sum_{k=1}^K \left((\rho_n^2 + \vartheta_n^u R_{n,k}^u) - \vartheta_k^s R_{n,k}^u \right) \quad (25)$$

where $R_{n,k}^u$ is the achievable rate of subscriber n in Mbps mapped to slice k , ϑ_n^u is the unit price offered by MVSO to subscribers in \$/Mbps. We assume each MVSO represents a slice k , and multiple users N are admitted into slice K .

The objective function is to maximize variables ρ_n and ϑ_n^u , while considering constraints. This is a mixed integer nonlinear programming (MINLP) optimization sub-problem.

$$\underset{\rho_n, \vartheta_n^u}{\text{maximize}} \quad \bar{F}_m \quad (26)$$

subject to $C_7 - C_{11}$

The first constraint C_7 ensures that the bandwidth B_n^u in MHz allocated to UT n does not exceed the total available leased bandwidth B_k^s in MHz for slice K .

$$C_7 : \sum_{n=1}^N B_n^u \leq \sum_{k=1}^K \alpha_k B_k^s \quad \forall n \in N \quad (27)$$

Constraint C_8 ensures that the user GBR R_n^u for all users N by the MVSO is less than or equal to the GBR of the slice

R_k^s offered by the InP in the SLA. The eMBB users may use MC to achieve the GBR.

$$C_8 : \sum_{n=1}^N R_{n,k}^u \leq \sum_{k=1}^K \alpha_k R_k^s \quad \forall k \in K \quad (28)$$

Constraint C_9 , inspired from [45], ensures ρ_n increases as $R_{n,k}^u$ increases, and reduces when N_k^u or ϑ_n^u or the delay of the user $D_{n,k}^u$ increases. This constraint is vital as it captures the subscriber's quality of experience (QoE), which depends on the variables mentioned above, impacting the MVSO revenue, especially in the long term.

$$C_9 : \rho_n = \frac{R_{n,k}^u}{\left(N_k^u + \vartheta_n^u + D_{n,k}^u \right)} \quad \forall n \in N \quad (29)$$

To enable the MVSO to profit from the leased bandwidth capacity of the InP, constraint C_{10} ensures that the unit price ϑ_n^u set by the MVSO to the user is greater than the unit price ϑ_k^s set by the InP to the MVSO.

$$C_{10} : \vartheta_k^s < \vartheta_n^u \quad (30)$$

Constraint C_{11} ensures that the delay for user n matches the delay of the slice k it is mapped to.

$$C_{11} : D_{n,k}^u = D_k \quad \forall n \in N \quad \forall k \in K \quad (31)$$

All the constraints in this Section are equality and inequality constraints.

C. Multi-Objective Problem Formation

The two objectives in (20) and (18) are combined to form the MOOP in (26), which is solved in Section IV.

$$\underset{\vartheta_k^s, \beta_{k,s}, \rho_n, \vartheta_n^u}{\text{max}} \quad (F_i, \bar{F}_m) \quad (32)$$

subject to $C_1 - C_{11}$.

The variables common to F_i and $\bar{F}_m$ are R_k^s , B_k^s , α_k , D_k and ϑ_k^s . The variables unique to F_i are $\beta_{k,s}$ and $\bar{B}_s$, while those unique to $\bar{F}_m$ are ρ_n , ϑ_n^u , B_n^u , $R_{n,k}^u$ and $D_{n,k}^u$.

IV. PROPOSED SOLUTION STRATEGY FOR MOOP

The revenue maximization with efficient resource utilization by network slicing for the two objective functions $\bar{F}_m$ and F_i is a MOOP along with the equality and inequality constraints C_1 to C_{11} . Each objective problem function involves maximizing operations with respect to their respective variables ϑ_k^s , $\beta_{k,s}$, ρ_n and ϑ_n^u . These are also combinatorial problems, as $\bar{F}_m$ evaluates to determine combinations of variables, including the slice and satellite allocation (K and S), while F_i similarly searches for combinations of the user and satellite association (N and K) variables.

Solving the MOOP in (32) poses a significant challenge. Therefore, the solution strategy is subdivided into three parts: pricing mechanism, AC, and traffic prediction. Each of these components is addressed using different techniques and algorithms to arrive at a comprehensive solution.

A. Pricing Mechanism Solution

The pricing optimization within the MOOP is addressed through a comprehensive strategy involving three distinct approaches, each outlined below. These approaches include the utilization of advanced algorithms such as non-dominated sorting genetic algorithm, multi-objective reinforcement learning approach, and a developed heuristic approach discussed in Section V-B1. Each of these methods brings unique advantages to the optimization process, contributing to a robust solution framework.

1) *Non-Dominated Sorting Genetic Algorithm II*: The pricing aspect of the MOOP problem for joint NTN slicing and AC can be solved using the NSGA-II algorithm. The objective of solving (32) is to maximise ϑ_k^s , $\beta_{k,s}$, ρ_n and ϑ_n^u . The NSGA-II algorithm is considered due to its comparatively lower complexity, faster convergence, scalability, and ability to provide solutions that are very close to optimal, as compared to other solvers or algorithms for MOOPs [46]. The following will discuss the methodology of NSGA-II algorithm from [47], [48], [49]; firstly technical parameters will be explained for better understanding.

An individual or genome is any point to which a fitness function can be applied. A fitness function is a measure intended to be optimized. An array of individuals is considered a population or generation, and the average distance between individuals that make up a population is known as diversity, an important factor in the algorithm as it allows for search over a certain area. A fitness value is the value of the fitness function for a particular individual. The algorithm creates the next generation by selecting special individuals with the best fitness values in the current population, known as parents, which are used to create the next generation of individuals referred to as children. An individual point r can dominate another point v when the following conditions are satisfied in a vector-valued function f .

$$f_i(r) \leq f_i(v) \forall i; \quad f_j(r) < f_j(v) \text{ for some } j \quad (33)$$

A non-dominated set of points consist of points that do not have any other point dominating them. The algorithm uses *rank* to select the parents, with the rank of an individual indicating its level of dominance. Individuals with rank 1 are not dominated by any other individuals, while rank 2 individuals are dominated only by those in rank 1. Further, crowding distance (Dis) is a technique used to evaluate the distance between an individual and its closest neighbour. The NSGA-II algorithm evaluates this distance in a decision variable space x for individuals of similar rank. This distance is measured as a summation over the normalized distances between the individuals i and their corresponding neighbours within a sorted dimension (q).

$$Dis(i) = \sum_q (x(q, i+1) - x(q, i-1)) \forall q \quad (34)$$

Spread is a measure of the extent to which the Pareto set moves. It is calculated by first computing the standard deviation (δ) of the crowding distance measurement for the

points on the Pareto front with finite distances. The spread can be evaluated as follows:

$$spread = \frac{\varphi + \delta}{\varphi + N_p \Delta} \quad (35)$$

where N_p is the number of points, Δ is the average measure of distance between the points, and φ is the calculated summation across the objective function indices of the norm. This norm represents the difference between the current minimum Pareto point associated with that index and the corresponding minimum point from the previous iteration.

Hence, the NSGA-II implemented here follows the procedure outlined below. The chromosome (solution) structure in the NSGA-II, as it relates to this problem, includes the following variables: ϑ_k^s , ϑ_n^u , ρ , and $\beta_{k,s}$. The algorithm starts by generating the initial population and determining the number of the individuals. In this problem, the number of population is $[(N \times K) + (S \times K)]$. This population is generated based on feasibility with respect to upper/lower bounds and the constraints given in $C_1 - C_{11}$. Next, the population is sorted by non-domination, and each individual is assigned a rank/fitness corresponding to the non-dominated level. The fitness function in this case is (32). Crowding distance analysis is performed for the individuals using (34), ensuring diversity is maintained in the population. Parent selection occurs, constituting the next generation from the current population using a combination of crowding distance and ranking. Subsequently, crossover and mutation operations are performed on the parents to create the new generation of children. The encoding process used for chromosome creation allows the algorithm to explore different combinations through selection, crossover, and mutation to obtain the optimal solution.

2) *Multi-Objective Reinforcement Learning Approach*: Reinforcement learning (RL) is a form of ML algorithm that learns by interacting with the environment on a trial and error basis. The RL agent searches for actions based on set goals that optimize the reward while changing states to satisfy predefined goals [50]. When there are multiple objective functions in a related MOOP, the RL application in that scenario becomes a MORL.

- **Multi-Objective Markov Decision Process**: A RL problem is modelled as a Markov Decision Process (MDP) involving sequential decision-making, where the agent chooses actions to transition from one state to another, operating with uncertain probabilities and rewards. The outcome of actions taken is the reward. In MORL, the MDP becomes a multi-objective MDP (MO-MDP), where the single reward vector transforms into multi-objective reward vectors [25]. The MO-MDP of the MORL can be represented by a tuple $(S'_{s'}, A_{s'}, P_{s'}(S'_{t+1}|S'_t, a_t), R_{s'})$, where $S'_{s'}$ is the set of states, $A_{s'}$ is the set of actions taken by the agents InP and MVSO, $P_{s'}(S'_{t+1}|S'_t, a_t)$ is the set of transition probabilities after actions $A_{s'}$ are executed in state $S'_{s'}$, and $R_{s'}$ represents the reward sets from the executed action $A_{s'}$ [13], [51].
- **Policy**: The policy adopted here is epsilon-greedy, which operates with epsilon ϵ initially as 1 to choose actions randomly (by exploration) with higher probability, especially

in early training phase, along with an epsilon-decay rate of 0.995 that decreases ϵ by the epsilon-decay rate after each action selection, leading to exploitation as ϵ reduces to choose best-learned actions that yields the maximum reward.

- **Q-Learning:** The RL algorithm adopted to solve this MOOP is the Q-Learning algorithm. In the Q-Learning algorithm, the agent chooses actions a_t at time t in state S'_t to transition to the next state S'_{t+1} , and then receives a reward r_{t+1} from the environment [52], [53]. The state space is defined as $S'_t = \{1, 2, \dots, N_a\}$, where N_a is number of actions taken by the agent, and the environment is composed of the ρ_n , $R_{n,k}^u$, R_k^s , α_k , N , $\beta_{k,s}$ and ζ in this paper. In this MO-MDP, there are two agents for the InP and MVSO that take separate actions discretely, which involves choosing values of $\vartheta_k^s = \{1_{\vartheta_k^s}, 2_{\vartheta_k^s}, \dots, N_{\vartheta_k^s}\}$ and $\vartheta_n^u = \{1_{\vartheta_n^u}, 2_{\vartheta_n^u}, \dots, N_{\vartheta_n^u}\}$ respectively. Hence, the state space and the action space are similar, as implemented in [51]. The action function, as represented in [53], is given below:

For InP agent with Q-Table 1 (Q_1):

$$\text{action_}Q_1(t) = \begin{cases} \text{random action}(N_{\vartheta_k^s}), & \text{if probability } \epsilon \\ \text{argmax}(Q_1[S'_t, a_t]), & \text{probability } 1 - \epsilon \end{cases} \quad (36)$$

For MVSO Agent with Q-Table 2 (Q_2):

$$\text{action_}Q_2(t) = \begin{cases} \text{random action}(N_{\vartheta_n^u}), & \text{if probability } \epsilon \\ \text{argmax}(Q_2[S'_t, a_t]), & \text{probability } 1 - \epsilon \end{cases} \quad (37)$$

where $N_{\vartheta_k^s}$ and $N_{\vartheta_n^u}$ represent the number of ϑ_k^s and ϑ_n^u actions, respectively. This action function operates with two condition described in the policy above. The first is random action selection with probability ϵ , while the second condition is dependent on exploitation, selecting the maximum action from the Q-table. Further, the action selection by the agents leads to the result, and since the MOOP in this paper involves two objectives, the corresponding rewards are two (R_1 and R_2) presented below, which are equated to the objective function expressions of InP and MVSO respectively.

$$R_1 = \sum_{k=1}^K \sum_{s=1}^S \alpha_k \vartheta_k^s R_k^s \beta_{k,s} - \zeta \vartheta_k^s R_k^s \quad (38)$$

$$R_2 = \sum_{n=1}^N \sum_{k=1}^K \left((\rho_n^2 + \vartheta_n^u R_{n,k}^u) - \vartheta_k^s R_{n,k}^u \right) \quad (39)$$

The training of the agents is performed over episodes with a defined number of iterations. In order to maximize the Q-values of each state, the agents must select the appropriate action, based on an exploration-exploitation strategy for this work. During the testing phase, the agents exploit learned policies to choose actions without exploration. Hence, the Q-value function operates based on action selection in state so as to maximize the total reward in the environment [54]. The rewards are

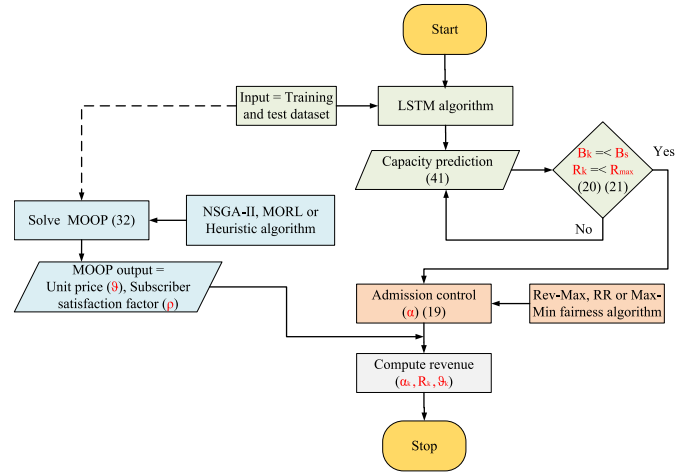


Fig. 4. The schematic flow design of proposed NAC-Rev-Max Algorithm.

obtained based on constraint satisfaction, especially C_{10} . Consequently, the Q-values for agents is updated using

$$Q(S'_t, a_t) = Q(S'_t, a_t)(1 - \delta) + \delta \left[r_{t+1} + \gamma \max_a Q(S'_{t+1}, a_t) \right] \quad (40)$$

where δ and γ are the learning rate and discount factor, respectively. The Q-table space is composed of rows and columns representing states and action, respectively. For this problem, the Q-table is a matrix of dimension $\{N_a \text{ by } N_{\vartheta_k^s}\}$ or $\{N_a \text{ by } N_{\vartheta_n^u}\}$ depending on the particular Q-table. Since we have two agents in this MO-MDP scenario, two Q tables (Q_1 and Q_2) are required, one for each agent.

B. The Admission Control Mechanism

We consider a proposed AC algorithm, explained in this section. The *revenue maximization (Rev-Max) algorithm* is presented as a novel AC approach. Two other benchmark approaches are discussed in Section V-C, which include the *max-min fairness algorithm*, and the *Round Robin approach*. Each of these approaches is designed to handle the acceptance of SRs with differing objectives and mechanisms.

1) *Admission Using Revenue Maximization Algorithm:* The Rev-Max algorithm operates by admitting SRs with a preference for high subscription traffic classes, specifically prioritizing eMBB traffic over URLLC in this paper. The eMBB traffic is considered and satisfied first before URLLC, as the objective is to maximize revenue by admitting as many premium slices as possible before the lower-premium ones.

C. Optimization of Capacity Prediction With Deep Learning

The objective is to minimize the loss function, specifically the MAE of the predicted traffic ($\widehat{R}_{k,p}^s$) as compared to the actual traffic ($R_{k,a}^s$). This objective is given as:

$$\min_{\theta_d} \frac{1}{T} \sum_{t=1}^T \left| R_{k,a}^s - \widehat{R}_{k,p}^s \right| \quad (41)$$

This loss function is optimized by modifying the neural network design and training parameters θ_d , which include batch size, epoch, neural network optimizer and activation type. The trade-off with these parameter changes is that they introduce additional complexity to the neural network. This is further discussed in Section V.

D. Proposed Algorithm

Our proposed algorithm, referred to as the Network Slicing with Admission Control for Revenue Maximization (NAC-Rev-Max) algorithm, consists of three distinct aspects presented by colors (green, blue, and orange) as depicted in Fig. 4. This flowchart illustrates the joint operation of network slicing and AC, aimed at maximizing revenue.

NAC-Rev-Max begins with traffic prediction, wherein the LSTM algorithm independently forecasts the eMBB and URLLC traffic after receiving and processing the dataset in according to the procedure outlined in Algorithm 3. These predicted traffic patterns provide the dynamic demand for eMBB and URLLC, incorporating variables S_w and R_k^s , which subsequently serve as essential inputs for the AC aspect highlighted in orange. This AC mechanism is designed to maximize revenue for both the InP and MVSO.

Following the prediction, NAC-Rev-Max addresses the pricing component of the MOOP using either NSGA-II (Algorithm 1), MORL (Algorithm 2) or the heuristic approach described in Section V-B1. The NSGA-II algorithm accepts input variables from the dataset features, defines the two objective functions along with the inequality constraints, and performs optimization to arrive at maximized values of unit prices ϑ_k^s and ϑ_n^u .

Alternatively, the MOOP pricing can be solved using the MORL Algorithm 2. This algorithm starts by initializing the environment and Q-agent parameters, then trains the agents over a defined number of time steps and episodes, accumulating rewards for every action taken by the agents as the state changes. The Q-values for both agents are updated using (40). During the testing phase, the procedure is performed over define time steps and episodes, with agents taking actions using the learned policies to obtain the resultant rewards. For the AC aspect, three algorithms are explored: the Max-Min fairness, round robin (RR) and the proposed Rev-Max AC algorithm. These three sub-algorithms are presented in Algorithm 4, where the syntax of the operation is outlined, and their objectives are described in Sections IV-B and V-C. Once the AC operation is completed and the unit price in \$/Mbps (from NSGA-II, MORL, or the heuristic approach) with the admitted slice capacity (predicted in part as demand by LSTM) in Mbps is derived, the revenue can be calculated. The revenue earned from the sliced network resources based on the dynamic traffic demand from all MVSOs is computed for the InP.

E. Complexity Analysis

The total complexity cost involved in solving problem (32) depends on the combined complexity of the NSGA-II, or

Algorithm 1: NSGA-II Algorithm to Obtain ϑ_k^s , ϑ_n^u and ρ_n

Input: Number of satellites = S ; Number of slices = K ; Number of User = N ; Satellite orbits & satellites limit = Ω ; Total available Bandwidth = B_s ; User bandwidth = B_n^u ; GBR range per user = $R_{n,k}^u$; GBR range per slice = R_k^s

- 1 **Call the functions to solve optimization**
- 2 *Solve the MOOP in (32) by calling functions below with NSGA-II to obtain ϑ_k^s , ϑ_n^u , ρ_n and $\beta_{k,s}$*
- 3 **Define InP function (17):**
- 4 **Function** Fi_InfraPro(*num_inputs*):
- 5 **for** $k \leq K$ **do**
- 6 **for** $s \leq S$ **do**
- 7 $F_i = \sum_{k=1}^K \sum_{s=1}^S \alpha_k \vartheta_k^s R_k^s \beta_{k,s} - \zeta \vartheta_k^s R_k^s$;
 // (F_i :To maximize ϑ_k^s and $\beta_{k,s}$)
- 8 **end**
- 9 **end**
- 10 **return** F_i
- 11 **Define MVSO function (25):**
- 12 **Function** Fm_MVSO(*num_inputs*):
- 13 **for** $n \leq N$ **do**
- 14 **for** $k \leq K$ **do**
- 15 $\bar{F}_m = \sum_{n=1}^N \sum_{k=1}^K ((\rho_n^2 + \vartheta_n^u R_{n,k}^u) - \vartheta_k^s R_{n,k}^u)$;
 // ($\bar{F}_m$:To maximize ϑ_n^u and ρ_n)
- 16 **end**
- 17 **end**
- 18 **return** F_2
- 19 **Define Constraint function for C_1 to C_{11} :**
- 20 **Function** constraint_ceg(*num_inputs*):
- 21 *constraints return Concatenation of constraints*
- 22 **End**

Algorithm 2: MORL With Q-Learning Multi-Agent Algorithm to Obtain ϑ_k^s and ϑ_n^u

- 1 **Initialize environment parameters:** $K, S, N, \zeta, \rho_n, \vartheta_k^s, \vartheta_n^u, \beta, \delta, \gamma, \epsilon, \epsilon_{decay}, max_episodes, max_episodes_test, max_step$;
- 2 **Initialize environment env with parameters;**
- 3 **Initialize the agent Q – agent with parameters;**
- 4 **Training Procedure:**
- 5 **for** $i \leftarrow 1$ **to** $max_episodes$ **do**
- 6 **Randomly select initial state** $state0$;
- 7 **for** $w \leftarrow 1$ **to** max_step **do**
- 8 **Choose actions** $action_Q1, action_Q2$ **using** **epsilon-greedy policy;**
- 9 **Receive rewards** $R1$ **and** $R2$ **by taking actions;**
- 10 **Update Q-values for both agents using (40);**
- 11 $Q(S_t^i, a_t) = Q(S_t^i, a_t)(1 - \delta) + \delta[r_{t+1} + \gamma \max_a Q(S_{t+1}^i, a)]$;
- 12 **Decay exploration rate** ϵ ;
- 13 **end**
- 14 **end**
- 15 **Testing Procedure:**
- 16 **for** $i \leftarrow 1$ **to** $max_episodes_test$ **do**
- 17 **Get the state from the environment** $state0 = env.get_state()$;
- 18 **for** $w \leftarrow 1$ **to** max_step **do**
- 19 **Choose actions** $action_Q1, action_Q2$ **using learned policies;**
- 20 **Receive rewards** $R1, R2$ **by taking actions while considering** $C_{10} [\vartheta_k^s < \vartheta_n^u]$;
- 21 **end**
- 22 **end**
- 23 **Export points for** $\vartheta_k^s, \vartheta_n^u$;
- 24 **Plot rewards results:** *Using matplotlib.pyplot*

MORL with Q-learning, LSTM, and AC algorithms. The initial non-dominated sorting genetic algorithm (NSGA) utilized a non-elitism approach, resulting in a high complexity of $O(MW^3)$, where M is the number of objective functions and

Algorithm 3: LSTM for Traffic Prediction of R_k^s and $R_{n,k}^u$

Data: Perform data preprocessing by separating the training and test set of the dataset.

- 1 *Import necessary Keras libraries and load the training set;*
- 2 **Perform feature scaling:** Scale using *MinMaxScaler* on training data of eMBB and URLLC;
- 3 **Build LSTM-based RNN model;**
- 4 **Define parameters:** Epochs = E_p ; batch-size = B_{size} ; activation function = *Relu*; optimizer = *Nadam*;
- 5 **for** $t \leq E_p$ **do**
- 6 **for** $q \leq B_{size}$ **do**
- 7 Extract a mini-batch of eMBB and URLLC data samples;
- 8 **Forward pass;**
- 9 Pass the mini-batch through the LSTM network;
- 10 Calculate the predicted traffic probabilities;
- 11 **Backward pass;**
- 12 Compute the prediction error with mean-absolute-error loss function (41);
- 13 Update LSTM weights using backpropagation through time (BPTT);
- 14 **end**
- 15 **end**
- 16 **Prediction:** Import test-set and make prediction using the trained RNN model;
- 17 **Display results:** Visualize the results for eMBB and URLLC using *matplotlib.pyplot*

Algorithm 4: Admission Control With Rev-Max, Round Robin and Max-Min Fairness

Input: Total satellite capacity(GEO/MEO/LEO) = $Total_C$; URLLC Demand = $urllc_demand$; eMBB Demand = $embb_demand$; URLLC $S_w = urllc_slice_weight$; eMBB $S_w = embb_slice_weight$;

- 1 **Call the AC functions**
- (*Rev - Max result output*) $\leftarrow$ *Rev_max_allocation(input)*
- (*Max - Min fairness result output*) $\leftarrow$ *Max_min_fairness_allocation(input)*
- (*RR result output*) $\leftarrow$ *Round_Robin_allocation(input)*
- 2 **Function** *Rev_Max_allocation(inputs)*:
- 3 **for** $embb_demand \leq Total_C$ **do**
- 4 First allocate capacity to highest priority service (which is eMBB) based on S_w and available capacities (starting with GEO $\rightarrow$ MEO $\rightarrow$ LEO)
- 5 **end**
- 6 **for** $urllc_slice_weight \leq (Total_C - embb_demand)$ **do**
- 7 Allocate remaining capacity (of LEO) to URLLC based on S_w
- 8 **end**
- 9 **return** *Rev_max_urllc*, *Rev_max_embb*, *Rev_max_un_allocated_capacity*
- 11 **Function** *max_min_fairness_allocation(inputs)*:
- 12 Ensure that minimum demand (URLLC) is satisfied first based on S_w and available capacities (starting with LEO $\rightarrow$ MEO $\rightarrow$ GEO). Then allocate remaining capacity to eMBB based on S_w
- 13 **return** *Max_min_urllc*, *Max_min_embb*, *Max_min_un_allocated_capacity*
- 15 **Function** *Round_Robin_allocation(inputs)*:
- 16 Allocate resources to eMBB and URLLC SR in sequential order without preference, until available resources is utilized, while considering the S_w .
- 17 **return** *RR_urllc*, *RR_embb*, *RR_un_allocated_capacity*
- 18 **End**
- 19 **Display allocation results**
- print(Allocation; Un_allocated; Un_met_demand; Penalty)*

W represents the population size. However, in this work, a faster non-dominated sorting mechanism known as NSGA-II is used. NSGA-II converges faster, employs an elitist approach with reduced complexity, and finds a better spread of solutions near the Pareto-optimal front. The complexity of NSGA-II in solving the MOOP is $O(MW^2)$ [47], where M is two in this

TABLE II
SIMULATION PARAMETERS

Items	Parameter at 17:00hr	Parameter at 2:00hr
URLLC S_w (Mbps)	400	350
eMBB S_w (Mbps)	930	730
URLLC demand (Mbps)	1,600	1,400
eMBB demand (Mbps)	4,650	3,650
URLLC maximum $R_{n,k}^u$ (Mbps)	47	47
eMBB maximum $R_{n,k}^u$ (Mbps)	115	115
URLLC B_n^u @ 2bps/Hz (MHz)	24	24
eMBB B_n^u @ 2bps/Hz (MHz)	58	58
Total available capacity (Mbps)	5,100	4,300
LEO percentage of total capacity (%)	20	20
MEO percentage of total capacity (%)	40	40
GEO percentage of total capacity (%)	40	40

paper, and W , the population size is equal to $[(N \times K) + (S \times K)]$ as discussed in Section IV-A. Hence, as M and W increase, the execution time and complexity of the NSGA-II also increase.

For MORL with Q-learning, the complexity of the algorithm depends on several parameters, such as the size of action space and the length of the time step (T_s) in the defined max episode (M_e). Since there are two agents in Algorithm 2, the action space is of size $N_{\vartheta_k^s}$ for the InP agent and $N_{\vartheta_n^u}$ for the MVSO agent. With L as the number of multiplication operations in the neural network, it can be assumed that the complexity of training for one experience is $O(L)$. Hence, the complexity of this MORL is $O(LT_s[N_{\vartheta_k^s} + N_{\vartheta_n^u}]M_e)$ [55]. Further, the complexity of LSTM per weight and time step can be represented as $O(W_t)$, where W_t is the weight/length of the input sequence [31]. LSTM complexity can also be derived, following [33], in terms of real value operations performed by the four gates that depend on the LSTM hidden state size Z and the input sequence length W_t . Each gate implements $3Z + W_t - 2$ summation and $Z^2 + ZW_t$ multiplication, resulting in the overall LSTM cost:

$$L_{Cost} = 4(Z^2 + ZW_t + 3Z + W_t - 2) + 4Z \quad (42)$$

The complexity of the AC algorithms, as described in [56], for the proposed priority-based Rev-Max algorithm results in an overall complexity derived from the SR acceptance feasibility checking based on available resources. The complexity of the admission is $O(K \times S) + \sum_{k=1}^K O(S^3) = O(S^3 + KS)$. Hence, the complexity increases as K (number of SRs) and S (number of satellites impacting capacity) increase.

V. PERFORMANCE EVALUATION

In this section, the simulation configurations and the performance analysis of the proposed algorithm are discussed, and evaluated against other state-of-the-art algorithms.

A. Simulation Setup

The simulation is first setup to solve the pricing strategy of the MOOP described in Section III-C using NSGA-II, MORL and the heuristic approach. The NSGA-II algorithm is configured in the MATLAB simulation environment along with the Heuristic approach, while the MORL algorithm is configured in Python using the same values of ρ_n as in NSGA-II. The system used runs Windows 10 with 32GB of RAM and 16 x Intel 2.40GHz cores. Specifically, Python 3.9.18 and MATLAB R2023b were employed for the simulations. The heuristic approach mentioned in Section V-B1 is simulated, where the required iterated value of ρ_n is adopted from that

of NSGA-II to create a basis for comparison. Similarly, for MORL algorithm, the same values of R_n^u , R_k^s , and ρ as in NSGA-II are used for the basis of comparison. In this paper, the number of actions, N_a , for each agent is configured as 15, and the maximum defined values of ϑ_k^s and ϑ_n^u for the MORL agents of InP and MVSO, are 9 \$/Mbps and 10 \$/Mbps, respectively. Precisely, the InP agent is configured to generate N_a number of values between $\frac{\vartheta_k^s}{N_a}$ and ϑ_k^s , leading to the actual range of discretize values of $\{0.6, 1.2, 1.8, 2.4, 3, 3.6, 4.2, 4.8, 5.4, 6, 6.6, 7.2, 7.8, 8.4, 9.0 \text{ $/Mbps}\}$. Similarly, the MVSO agent is configured to generate N_a number of values between $\frac{\vartheta_n^u}{N_a}$ and $N_{\vartheta_n^u}$, leading to the actual range of discretize values of $\{0.67, 1.33, 2.00, 2.67, 3.33, 4.00, 4.67, 5.33, 6.00, 6.67, 7.33, 8.00, 8.67, 9.33, 10.00 \text{ $/Mbps}\}$.

Further, the LSTM traffic prediction of eMBB and URLLC is simulated using python development environment in the Anaconda navigator suite, with configurations implemented with the keras library; the dataset pre-processing is done using MS Excel. For the traffic prediction optimization described in Section IV-C, the neural network is designed by considering different parameters to arrive at a suitable configuration with minimal loss function.

In Fig. 5 (a), the LSTM loss is evaluated with different deep learning optimizer including NADAM, ADAM, RMSPROP and SGD. The performance shows that NADAM achieved a loss of 0.0894 at epoch 100, while ADAM, RMSPROP, and SGD performed at 0.0900, 0.0908, and 0.0917, respectively. This indicates that NADAM offers the best performance over ADAM, RMSPROP, and SGD by 0.67%, 1.55%, and 2.54%, respectively. Further, the performance of loss with respect to epoch is shown in Fig. 5 (b). When different activation functions are considered, Relu shows the best performance with a value of 0.0894 at 100 epochs, while softplus and softmax are 0.0917 and 0.0922, respectively, indicating that Relu outperformed softplus and softmax by 2.54% and 3.08%, respectively. Similarly, when different batch sizes were considered in Fig. 5 (c), batch size 32 offers the best loss performance of 0.0894 at 100 epochs, while batch sizes 64 and 128 performed at 0.0927 and 0.0955, respectively, showing that batch size 32 outperforms batch sizes 64 and 128 by 3.62% and 6.59%, respectively. Hence, the LSTM network is configured using NADAM, Relu and batch size 32, resulting in the predicted eMBB and URLLC traffic shown in Fig. 5 (d). The predicted traffic shows a varying daily trend, with lower traffic from 1:00 hr to 6:00 hr, indicating less demand during these idle hours. The traffic trend at 2:00hr and 17:00hr is considered for low and high demand periods in this analysis, using 1,600 Mbps and 4,650 Mbps as the traffic demands from the prediction at 17:00hrs for URLLC and eMBB, respectively. The simulation is setup as mention above for the proposed algorithm using the parameters outlined in Table II, mostly derived from the dataset and LSTM traffic prediction, and this is compared with other state-of-the-art algorithms in terms of unit price, capacity allocation, revenue and peak delay.

B. Pricing Mechanism Benchmark

1) *Heuristic Approach*: This approach obtains ϑ_k^s using (46), while assuming ρ . The exercise is then

subsequently repeated iteratively for different values of ρ_n until a suitable solution for ϑ_k^s is obtained. The details of the derivation of ϑ_k^s are presented as follows. The Heuristic approach for solving ϑ_k^s is provided along with the details of the mathematical derivation, starting from constraint C_9 . Recall that ρ_n in (29) is given as

$$\rho_n = \frac{R_{n,k}^u}{(N_k^u + \vartheta_n^u + D_{n,k}^u)}$$

If satisfaction index (ρ_n) is assumed and other related variables are known, then ϑ_n^u can be obtained. Making ϑ_n^u the subject in (29), multiply both sides by $(N_k^u + \vartheta_n^u + D_{n,k}^u)$, which yields

$$\rho_n \cdot (N_k^u + \vartheta_n^u + D_{n,k}^u) = R_{n,k}^u \quad (43)$$

Expand the right and subtract $\rho_n \cdot \vartheta_n^u$ from both sides.

$$\rho_n \cdot \vartheta_n^u = R_{n,k}^u - \rho_n \cdot N_k^u - \rho_n \cdot D_{n,k}^u \quad (44)$$

Then ϑ_n^u is obtained as shown below:

$$\vartheta_n^u = \frac{R_{n,k}^u - \rho_n \cdot N_k^u - \rho_n \cdot D_{n,k}^u}{\rho_n} * N_f \quad (45)$$

Note: A normalization factor N_f is included to normalize the resultant price accordingly. Further, recalling constraint C_{10} , which states $\vartheta_k^s < \vartheta_n^u$, this leads to the final outcome of

$$\vartheta_k^s < \frac{R_{n,k}^u - \rho_n \cdot N_k^u - \rho_n \cdot D_{n,k}^u}{\rho_n} * N_f \quad (46)$$

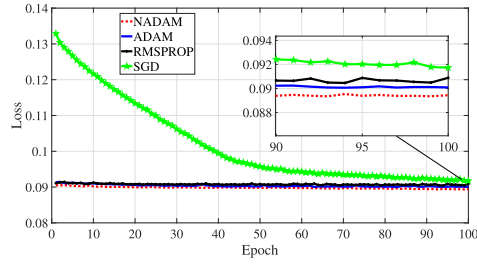
C. Admission Control Benchmark

1) *Admission Using Max-Min Fairness Algorithm*: The max-min fairness algorithm operates by admitting SRs with the minimum demand first, and subsequently admitting demands from other traffic classes using a similar proportion based on the number of accepted SRs while considering the remaining available resources [57]. The objective is to fairly admit as many SRs, irrespective of the traffic type, in an equal manner.

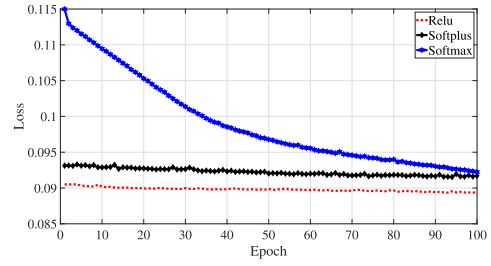
2) *Admission Using Round Robin Algorithm*: The RR algorithm admits SRs by allocating the resources of URLLC and eMBB demands in sequential order based on the S_w magnitude for each, admitting without bias and preference until available resource is exhausted [58]. A S_w is the unit of SR per individual MVSO request, out of the total SR in that traffic category. The objective is to admit as many S_w units of the SRs sequentially as they arrive the network.

D. Performance Analysis

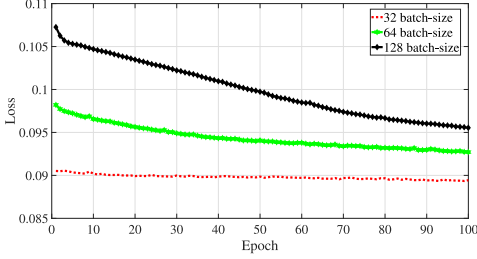
The objective of this work is to maximize revenue for the InP, which leases NTN resources dynamically through slicing to MVSOs with SLAs guaranteeing varying demanded QoS, to achieve high subscriber satisfaction and ensure efficient utilization of network resources. To achieve this objective, the unit prices of ϑ_k^s and ϑ_n^u in the MOOP are maximized for each of the objective functions in (18) and (26), respectively. Here, the InP and the MVSO achieve correspondingly satisfactory unit prices based on defined constraints. Specifically, the



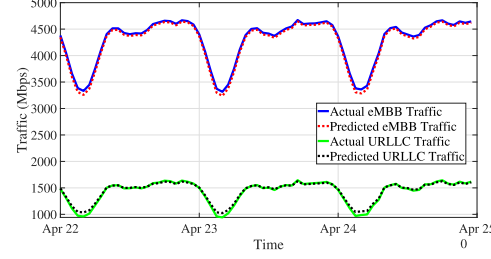
(a) Evaluation for the different deep learning optimizers.



(b) Evaluation for the different activation functions.

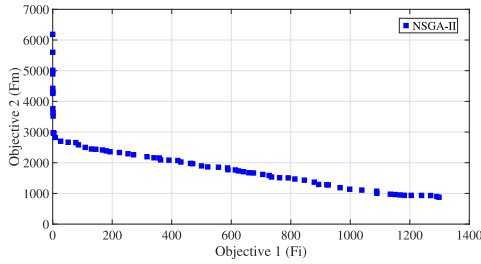


(c) Evaluation for the different batch sizes.

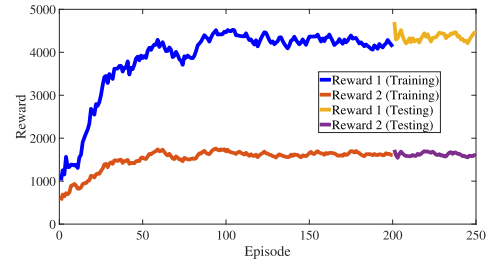


(d) Evaluation for the predicted eMBB and URLLC Traffic.

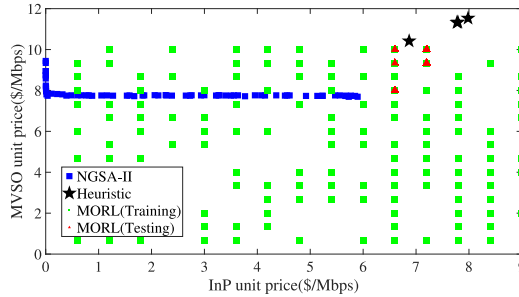
Fig. 5. Evaluation of LSTM RNN-model with respect to different optimizers, activation functions and batch sizes.



(a) Pareto front points from NSGA-II.



(b) MORL rewards analysis under training and testing phases.



(c) Resultant unit prices for NSGA-II, MORL and Heuristic.

Fig. 6. Evaluation of the Pareto front and resultant unit price for NSGA-II, MORL and Heuristic approach.

MVSO derives a unit price for the subscriber that can cover the cost of leasing from the InP and still make marginal profit, while the InP operates the leasing of resources with a commensurate compensation in the form of a defined unit price for the MVSO as a fee to cover the cost of resources offered. The InP and MVSO unit prices are obtained from optimization and bound by a condition stated in constraint c_{10} (30). This is done using three approaches: NSGA-II optimization, MORL with Q-Learning, and a Heuristic approach, where the Pareto points are obtain for ϑ_k^s and ϑ_n^u .

In Fig. 6 (a), the Pareto front points are shown for objective functions F_i and $\bar{F}_m$, with the resultant point for $\bar{F}_m$ mapped

onto that of F_i on the Pareto front while using the NSGA-II algorithm. Similarly, the MORL with Q-Learning algorithm is used to analyze the rewards R_1 and R_2 for F_i and $\bar{F}_m$, respectively, under training and testing phases, to evaluate the optimization performance in Fig. 6 (b).

Further, the optimization results of the unit prices of ϑ_k^s and ϑ_n^u in \$/Mbps are displayed in Fig. 6 (c) for the three approaches of NSGA-II, MORL and Heuristic. The NSGA-II optimization yields several unit prices, but what is considered are the optimized unit prices of 5.89 \$/Mbps and 7.69 \$/Mbps for ϑ_k^s and ϑ_n^u , respectively. This is achieved by considering the highest optimized unit price for the InP first and then

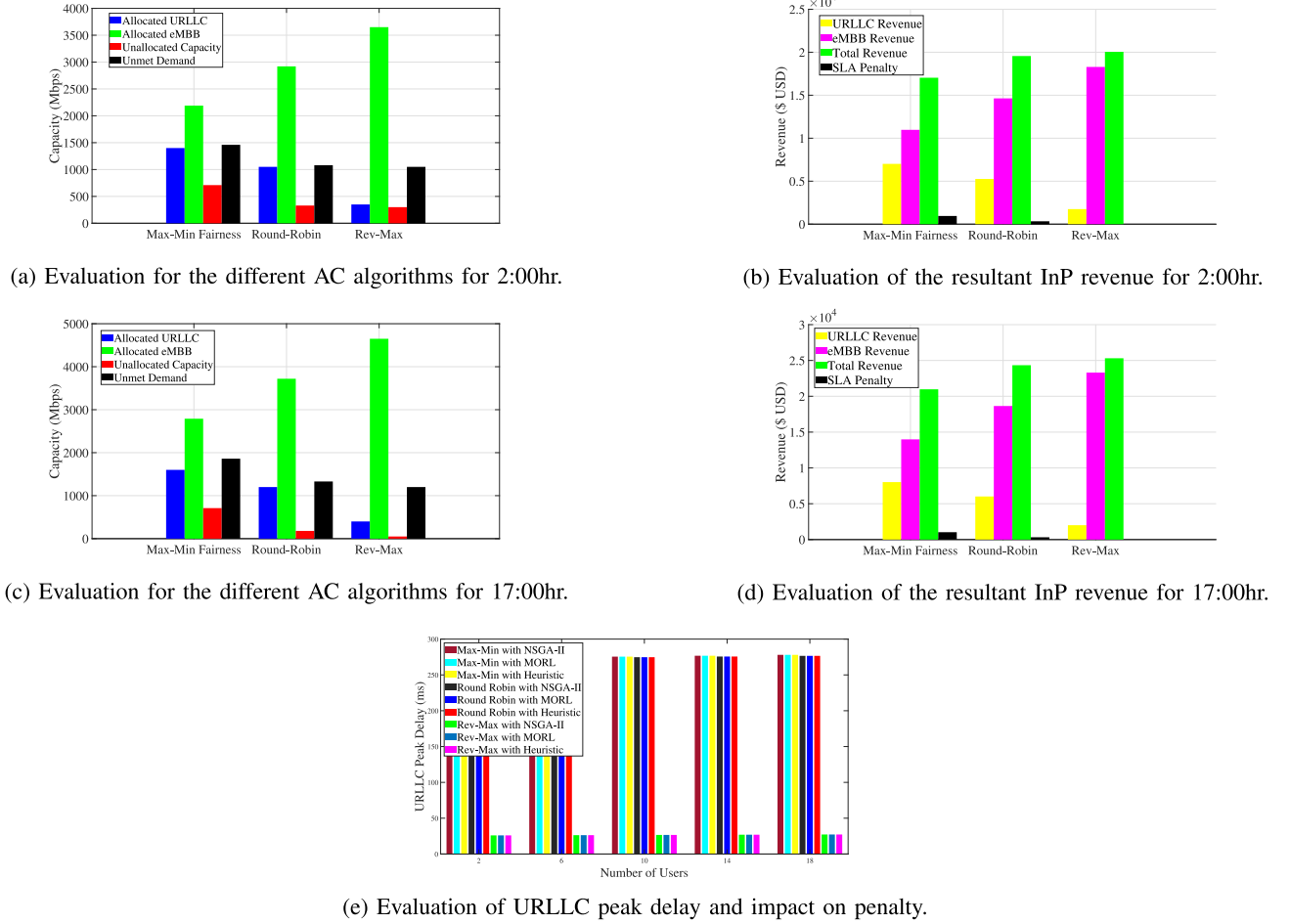


Fig. 7. Evaluation of cognitive behaviour of the admission control algorithms with respect to slice allocated capacity, un-allocation, unmet demand, achieved revenue and penalty imposed.

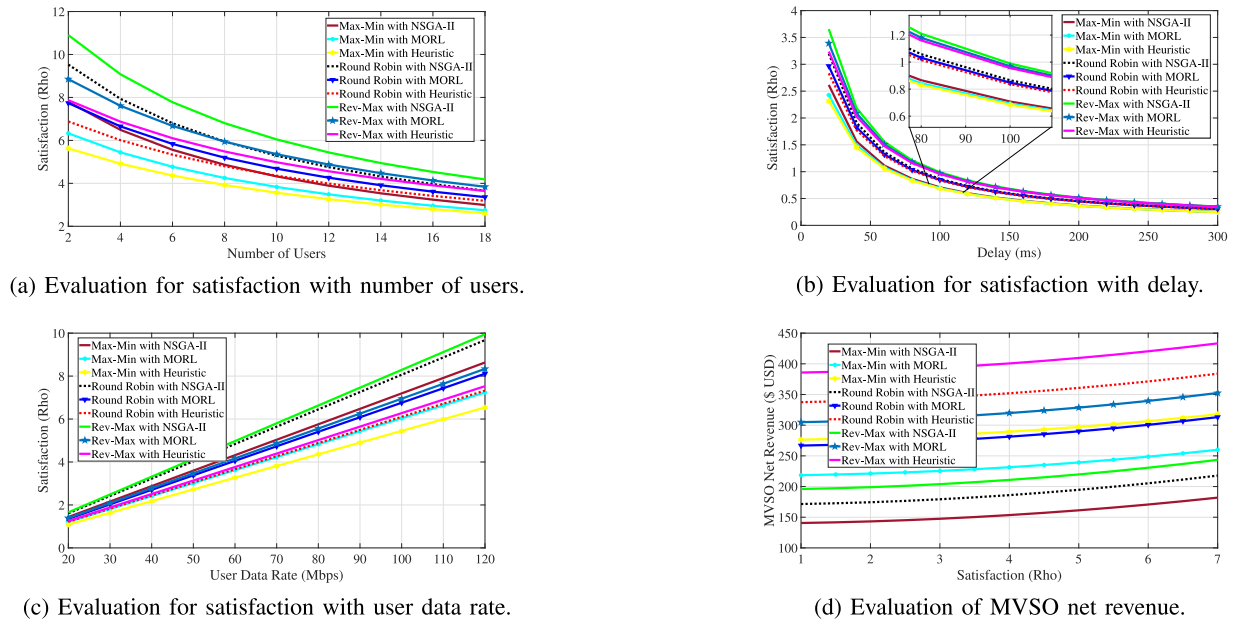


Fig. 8. Evaluation of the subscriber satisfaction index with respect to number of users, delay, data rate and MVSO net revenue for the different AC algorithms.

its corresponding MVSO pair. For MORL, the testing phase results are considered, resulting in different pairs of unit prices (ϑ_k^s and ϑ_n^u) based on the defined state space. The unit price

combination adopted for MORL is 7.19 \$/Mbp and 10.00 \$/Mbp for ϑ_k^s and ϑ_n^u , respectively, as this pair yields the highest optimized InP price. The Heuristic approach produces

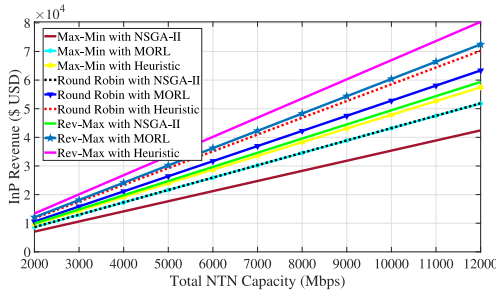


Fig. 9. Evaluation of impact of the leased network capacity on infrastructure provider net revenue.

a linear trend of different unit prices. Here, what is considered are the unit prices of 7.98 \$/Mbps and 11.52 \$/Mbps for ϑ_k^s and ϑ_n^u , respectively. The maximum value with reference from ϑ_k^s is selected and then mapped to the corresponding ϑ_n^u value. These results show that NSGA-II offers a more cost-effective price to the subscribers at 7.69 \$/Mbps compared to MORL at 9.33 \$/Mbps and the Heuristic at 11.52 \$/Mbps, representing a 19.27% improvement over MORL and a 39.87% improvement over the Heuristic.

Further, the three AC mechanisms adopted are the Max-Min fairness AC, RR AC, and the proposed Rev-Max AC, as outlined in Algorithm 4. In Fig. 7 (a), the AC performance is shown with respect to capacity allocation in Mbps for the lowest demand in the day at 2:00hr. The Max-Min fairness algorithm allocates 1,400 Mbps for URLLC and 2,190 Mbps for eMBB, indicating that all URLLC demands are met while only a portion of the eMBB demands are satisfied. Specifically, 710 Mbps remains unallocated from the total available 4,300 Mbps capacity. As the unallocated capacity of 710 Mbps falls short of the required eMBB S_w of 730 Mbps, it cannot be utilized for the eMBB slice and therefore remains unutilized. The chart also shows the unmet demand of 1,460 Mbps, which is mostly from the total eMBB demand of 3,650 Mbps. Similarly, at 2:00hr, the RR AC algorithm allocates 1,050 Mbps for URLLC and 2,920 Mbps for eMBB, leaving 330 Mbps unallocated and an unmet demand of 1,080 Mbps. In contrast, the proposed AC algorithm of Rev-Max shows eMBB allocation of 3,650 Mbps, which is the total eMBB demand, and URLLC allocation of 350 Mbps based on available capacity. The unallocated capacity is 300 Mbps, and the unmet demand of URLLC is 1,050 Mbps. This performance shows that Rev-Max admitted a cumulative of 4,000 Mbps slice traffic while RR and Max-Min admitted only 3,970 Mbps and 3,590 Mbps slice traffic, indicating Rev-Max admitted 0.75% and 10.80% more than RR and Max-Min algorithms, respectively. Further, Rev-Max has the lowest unmet demand, which is 2.81% and 32.66% less than RR and Max-Min, respectively, including the lowest unallocated capacity, which is 9.52% and 81.18% less than that of RR and Max-Min fairness, respectively. The reason for the unmet demand is that the available capacity is less than traffic demand, as shown in Table II. Hence, the capability of the AC algorithms is evaluated to ascertain the superior AC algorithm that yields the lowest unmet demand.

The InP revenue for the slices admitted at 2:00hr is shown in Fig. 7 (b). The Max-Min fairness AC yielded \$ 7,015.80 and \$ 10,974.70 from URLLC and eMBB slice traffic, respectively.

However, a penalty of \$ 955.09 awarded against the InP for URLLC slices allocated on MEO and GEO satellites because the capacity for LEO is only 20% of the total available capacity, and LEO is the only NTN layer that is used for URLLC services. This penalty is assumed as 30% of the income from the capacity wrongly allocated as per SLA, hence the total revenue earned by Max-Min fairness is \$ 17,035.40 after penalty deduction. Similarly, RR yields \$ 5,261.85 from URLLC and \$ 14,632.90 from eMBB, with a SLA violation penalty of \$ 336.05, resulting in a cumulative revenue of \$ 19,558.70 after penalty deduction. In the same figure, the performance of Rev-Max is seen to yield \$ 1,753.95 and \$ 18,291.20 for URLLC and eMBB, respectively, resulting in a cumulative revenue of \$ 20,045.10 without any penalty deductions since all slices were allocated to the correct orbital satellite as per SLA. Evidently, the Rev-Max achieved a significant revenue advantage over both RR and Max-Min, generating 2.45% more revenue than RR and 16.23% more than Max-Min. This is primarily due to Rev-Max's ability to avoid penalties and accommodate a higher volume of eMBB slice traffic.

Fig. 7 (c) presents the AC performance at 17:00hr with respect to capacity allocation in Mbps. Specifically, Max-Min fairness algorithm allocates 1,600 Mbps for URLLC and 2,790 Mbps for eMBB, indicating that all URLLC demand is satisfied, while only part of eMBB is satisfied, leaving the 710 Mbps unallocated from the total available 5,100 Mbps capacity. Since the available capacity of 710 Mbps is less than the required eMBB S_w of 930 Mbps, the remaining capacity cannot be used for eMBB slice traffic. The chart further reveals an unmet demand of 1,860 Mbps, mostly from the total eMBB demand of 4,650 Mbps. Similarly, at 17:00hr, RR AC algorithm allocates 1,200 Mbps for URLLC and 3,720 Mbps for eMBB, with unallocated capacity of 180 Mbps and unmet demand of 1,330 Mbps. The proposed AC algorithm of Rev-Max shows eMBB allocation of 4,650 Mbps, which is the total eMBB demand, and URLLC allocation of 400 Mbps based on available capacity. As the remaining unallocated capacity (50 Mbps) falls short of the required S_w for URLLC traffic (400 Mbps), it remains unused. The unmet demand of URLLC slices is 1,200 Mbps. This performance shows that Rev-Max admitted a cumulative total of 5,050 Mbps slice traffic while RR and Max-Min admitted only 4,920 Mbps and 4,390 Mbps slice traffic, respectively. This indicates Rev-Max admitted 2.60% and 13.98% more than RR and Max-Min algorithms, respectively. Rev-Max also has the lowest unmet demand, which is 10.27% and 43.13% less than RR and Max-Min, respectively, including the lowest unallocated capacity, which is 113.04% and 173.68% less than that of RR and Max-Min fairness, respectively.

The resultant InP revenue from the 17:00hr can be seen in Fig. 7 (d), where Max-Min fairness yields \$ 8,018.05 and \$ 13,981.50 from URLLC and eMBB slice allocation, respectively. A penalty of \$ 1025.84 is awarded against InP as a result of the wrongly allocated URLLC traffic on other orbits other than LEO as per SLA. This penalty is 30% of the income from the wrongly allocated URLLC slice traffic, resulting in a total revenue of \$ 20,973.70 after the penalty deductions. Similarly, RR yields \$ 6,013.54 from

URLLC and \$ 18,642.00 from eMBB, with a SLA violation penalty of \$ 318.36 resulting in a cumulative revenue of \$ 24,337.10 after penalty deduction. Additionally, Rev-Max AC yields \$ 2,004.51 and \$ 23,302.50 from admitted URLLC and eMBB slices, respectively, with a cumulative revenue of \$ 25,307 without any incurred penalty. This shows that Rev-Max outperformed RR and Max-Min by 3.91% and 18.73% in total earned revenue, respectively.

In Fig. 7 (e), the reason for the penalty on Max-Min and RR is confirmed, as NSGA-II, MORL, and Heuristic approaches for Max-Min fairness AC show a peak delay performance of 275.50 ms, and that of RR AC with NSGA-II, MORL and Heuristic is 274.72 ms when the number of users is 10 for URLLC slices. In contrast, Rev-Max with any of NSGA-II, MORL and Heuristic approach shows a peak delay performance of 26.61 ms with 10 users for URLLC admitted slices. Particularly, Rev-Max demonstrates superiority in URLLC peak delay with 164.67% and 164.76% improvements over RR and Max-Min fairness AC, respectively. This is achieved by intelligently limiting URLLC admissions below the available capacity and strategically allocating slices in a way that optimizes orbital resources while ensuring SLA compliance. Further, the impact of the subscriber satisfaction index is discussed with the dependant variables, which is important as the QoE can impact subscriber patronage and revenue for the MVSOs in the long term.

In Fig. 8 (a), the performance indicates that ρ decreases as the number of users increases. This observation aligns with our theoretical analysis presented in (29), where ρ_n is inversely proportional to the number of users due to the possibility of resource congestion. Additionally, Rev-Max with NSGA-II achieves ρ of 6.04 with 10 numbers of users, which is higher than Rev-Max with MORL at 5.35 and Rev-Max with Heuristic at 4.98 because Rev-Max with NSGA-II offers a lower unit price that is 26.11% and 39.87% less than the MORL and Heuristic. The trend also shows that at 10 numbers of users while analyzing for different AC algorithms operating the same pricing optimization, Rev-Max with NSGA-II achieves a higher ρ compared to RR with NSGA-II at 5.28 and Max-Min with NSGA-II at 4.32. Similarly, Rev-Max with MORL achieves higher ρ than RR with MORL at 4.68 and Max-Min with MORL at 3.83. Likewise, Rev-Max with Heuristic yields a higher ρ compared to RR with Heuristic at 4.35 and Max-Min with Heuristic at 3.56. This is because Rev-Max will always allocate slices to the appropriate orbital resources without violating SLA, whereas Max-Min violates SLA by allocating slices to wrong orbital satellites, especially URLLC slices, thereby impacting user satisfaction.

Similarly in Fig. 8 (b), it shows that as delay increases, the subscriber satisfaction reduces. At 100 ms delay, Rev-Max with NSGA-II shows the best ρ performance of 0.98, while Rev-Max with MORL and Heuristic are 0.96 and 0.95, respectively, because Rev-Max with MORL and Heuristic offers higher unit prices. The same trend applies to RR and Max-Min with their different pricing optimization algorithms. While analyzing across the different AC algorithms with the same pricing optimization algorithm, it can be observed that Rev-Max with NSGA-II achieves higher ρ at 100 ms compared

to RR with NSGA-II at 0.86 and Max-Min with NSGA-II at 0.70. The same trend applies to Rev-Max with MORL, which yields higher ρ at 100 ms than RR with MORL at 0.84 and Max-Min with MORL at 0.69. This is because RR and Max-Min allocate some URLLC slices to the wrong orbital satellites, potentially leading to SLA violations.

Fig. 8 (c) shows that as the data rate increases, sometimes with MC for eMBB traffic, the satisfaction of the users also increases because throughput is directly proportional to ρ in (29). At 90 Mbps while analyzing for different AC with the same pricing optimization strategy, Rev-Max with NSGA-II shows a higher performance of 7.46 ρ , while RR with NSGA-II yields 7.25 and Max-Min with NSGA-II achieves 6.47. This is because RR with NSGA-II and Max-Min with NSGA-II wrongly allocates resources to slices, which violates SLA, especially URLLC slices, while the former ensures all admitted slices are allocated to the right orbital satellites. The same trend applies to RR and Max-Min with their respective pricing strategies. Similarly, while analyzing performance across the same AC with different pricing strategy at 90 Mbps, it is observed that Rev-Max with NSGA-II yields a higher ρ compared to Rev-Max with MORL at 6.25 and Rev-Max with Heuristic at 5.64. The same trend applies to RR and Max-Min with the different pricing strategies. This is due to the lower unit price offered by NSGA-II compared to MORL and Heuristic algorithms. This confirms that Rev-Max with NSGA-II is superior in offering the highest user satisfaction, being 2.85%, 14.21%, 17.65%, 20.38%, 27.78%, 30.42%, 31.67% and 41.42% better than RR with NSGA-II, Max-Min with NSGA-II, Rev-Max with MORL, RR with MORL, Rev-Max Heuristic, RR with Heuristic, Max-Min with MORL and Max-Min with Heuristic, respectively.

In Fig. 8 (d), the performance of MVSO net revenue is displayed for all scenarios of admission control algorithms. When all AC scenarios operate with the same ρ of 5, the Rev-Max with Heuristic achieves \$ 409.54, while RR with Heuristic, Rev-Max with MORL, Max-Min with Heuristic, RR with MORL, Max-Min with MORL, Rev-Max with NSGA-II, RR with NSGA-II and Max-Min with NSGA-II achieve \$ 360.70, \$ 328.66, \$ 297.11, \$ 289.95, \$ 239.21, \$ 219.75, \$ 194.68 and \$ 161.24, respectively. This indicates that Rev-Max with Heuristic and RR with Heuristic are the top two performers because they admit more eMBB slices, which have higher S_w and subscription value than the URLLC slices. Consequently, Rev-Max with Heuristic is 60.31% higher than Rev-Max with NSGA-II due to the former's higher unit price. However, in the long term, the potential drawback of Rev-Max with Heuristic over Rev-Max with NSGA-II is that subscribers might gradually migrate to the latter since it offers the same QoS/QoE at a moderate/lower unit price.

Finally, in Fig. 9 the performance of InP revenue is assessed for all AC algorithms. It reveals that as the available NTN capacity for lease to MVSOs increases, the InP earns more revenue. Rev-Max with Heuristic achieves \$ 66,951.60 when leased NTN capacity is 10,000 Mbps, while the Rev-Max with MORL, RR with Heuristic, RR with MORL, Rev-Max with NSGA-II, Max-Min with Heuristic, RR with NSGA-II, Max-Min with MORL and Max-Min NSGA-II achieve \$ 60,387.00, \$ 58,539.00, \$ 52,799.30, \$ 49,447.10, \$ 47,877.70,

\$ 43,234.00, \$ 43,183.30, and \$ 35,360.10, respectively at 10,000 Mbps capacity lease. This demonstrates that Rev-Max with Heuristic outperforms Rev-Max with MORL, RR with Heuristic, and Rev-Max with NSGA-II algorithms by 10.31%, 13.41% and 30.07%, respectively, because it admits more of eMBB slices with bigger S_w and higher subscription value than other configurations and offers a higher unit price ϑ_s^k than the other pricing algorithms.

Accordingly, it can be confirmed that Rev-Max AC with Heuristic algorithm yields higher revenue for InP compared to the other algorithms. Similarly, in terms of the MSVOs' net revenue, Rev-Max with Heuristic algorithm offers the highest value when compared to the other algorithms. While Rev-Max with a Heuristic approach initially delivers a higher net revenue compared to Rev-Max with NSGA-II, the latter might offer sustainable long-term benefits through higher subscriber satisfaction. This is because Rev-Max with NSGA-II prioritizes a moderate unit price while maintaining the same QoS/QoE and adhering to SLAs. This strategy can enhance customer loyalty and potentially lead to more stable revenue growth in the long run compared to the potentially short-term gains of the Rev-Max with Heuristic algorithm.

VI. CONCLUSION

This paper presents a model of integrated service delivery, wherein users subscribe to MVSOs who lease NTN infrastructure from satellite InPs. A novel NTN slicing architecture was proposed, incorporating multi-layer satellites, including LEO, MEO, and GEO constellations, with an HGS adapted to the 3GPP NFV reference architecture. To this end, an MOOP was formulated for InPs and MVSOs, aiming at revenue maximization while minimizing SLA violations and ensuring QoS guarantees for subscribers. The solution to the MOOP is obtained in three parts, where the MOOP pricing optimization part was solved using different approaches: NSGA-II algorithm, MORL, and a heuristic process. These approaches yielded maximized variable vectors for the MOOP, represented as Pareto front points. The proposed algorithm (NAC-Rev-Max) has addressed the joint slicing and AC requirements through three sub-algorithms: unit price optimization, LSTM-based traffic prediction, and AC algorithm (Rev-Max). Specifically, NAC-Rev-Max prioritized admission of high-value slices, i.e., eMBB, over slices with lower subscription value, that is URLLC. Additionally, it intelligently allocated dynamic slices based on LSTM traffic prediction information.

Our proposed algorithm outperformed state-of-the-art algorithms (RR and Max-Min fairness approaches) in terms of subscriber satisfaction index, throughput, peak delay, MVSO revenue, and InP revenue. Precisely, in terms of peak revenue, the proposed algorithm outperformed RR and Max-Min fairness approaches by 3.91% and 18.73%, respectively. In short, our proposed model of integrated service delivery contributes to revolutionizing telecommunications ecosystems by empowering users to seamlessly access advanced services through leveraging satellite infrastructure, thereby enhancing connectivity and expanding network coverage in remote or underserved areas.

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